

# Results

01 • FISHER'S EXACT

exact test for small categorical tables

Table 1. Contingency

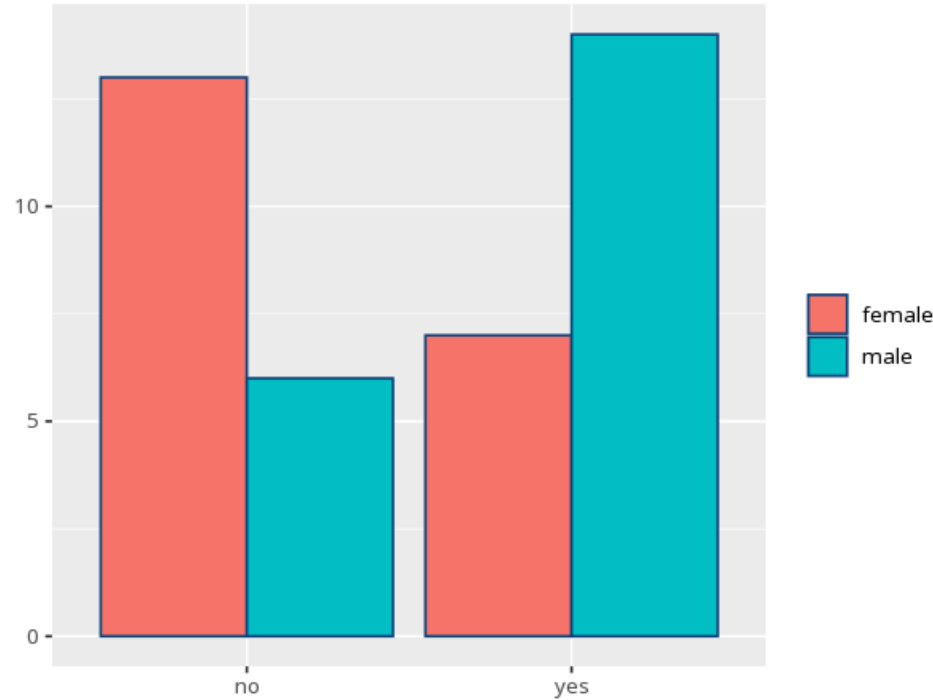
| Row \ Column | female | male | Total |
|--------------|--------|------|-------|
| no           | 13     | 6    | 19    |
| yes          | 7      | 14   | 21    |
| Total        | 20     | 20   | 40    |

Table 2. Fisher's exact test

| p (exact) | Odds ratio (cond. MLE) | 95% CI        | Cramér's V [95% CI] |
|-----------|------------------------|---------------|---------------------|
| .056      | 4.16                   | [0.97, 20.18] | —                   |

the odds ratio is fisher.test()'s conditional MLE (not the sample cross-product) and, with its CI, is reported for 2x2 tables only; for larger tables the OR is undefined, so Cramér's V (effectsize::cramers\_v) reports the strength of association alongside the exact p.

Figure. Cross-classification



How to read this test

An exact alternative to chi-square for small samples. A p below alpha means the two categories are associated; for 2x2 tables the odds ratio quantifies the association — this is fisher.test()'s conditional MLE (the non-central hypergeometric estimate), not the sample cross-product. OR>1 means the outcome is

more likely in the first row/group,  $OR < 1$  less likely,  $OR = 1$  no association; check the row/column order before reading direction, since the OR reflects the table's orientation. For tables larger than  $2 \times 2$  the odds ratio is undefined, so Cramér's  $V$  (0–1) reports the strength of association instead. Your significance threshold ( $\alpha$ ) is 0.05.

**APA template:** A Fisher's exact test of passed by gender gave  $p = .056$  ( $OR = 4.16$ , 95% CI [0.97, 20.18]).

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